

30 A proton and an alpha particle (${}^4_2\text{He}$ Helium nucleus) have the same de Broglie wavelength.

The ratio $\frac{\text{speed of proton}}{\text{speed of alpha particle}}$ is

A $\frac{1}{4}$

B $\frac{1}{2}$

C 2

D 4

Ans: D

By De Broglie's relation,

$$\frac{p_{\text{proton}}}{p_{\alpha}} = \frac{\lambda_{\alpha}}{\lambda_{\text{proton}}}$$

$$\frac{m_{\text{proton}}v_{\text{proton}}}{m_{\alpha}v_{\alpha}} = \frac{\lambda_{\alpha}}{\lambda_{\text{proton}}}$$

$$\frac{(u)v_{\text{proton}}}{(4u)v_{\alpha}} = \frac{1}{1}$$

$$\frac{v_{\text{proton}}}{v_{\alpha}} = 4$$

End of Paper